

Exercise 4.6

$$\Sigma = \sigma \mathbb{1} + i \tau \cdot \pi$$

$$\tau = (\sigma_1, \sigma_2, \sigma_3), \quad \pi = (\pi_1, \pi_2, \pi_3)$$

Σ transforms under the $SU(2)_L \times SU(2)_R$ symmetry as

$$\Sigma \rightarrow U_L \Sigma U_R^\dagger \quad \text{where} \quad U_{L,R} = \exp\left(-i \alpha_{L,R} \cdot \frac{\tau}{2}\right) \quad \text{are}$$

arbitrary $SU(2)$ matrices.

The mass dimensions are $[\sigma] = [\pi] = 1 \Rightarrow [\Sigma] = 1$

The Lagrangian is a scalar so we need to take trace of whatever matrix products we construct out of Σ .

$[\Sigma^\dagger \Sigma] = 2$ so if the symmetry transformation leaves $\text{tr}(\Sigma^\dagger \Sigma)$ invariant then this term can be included in the Lagrangian. We start with $\text{tr}(\Sigma^\dagger \Sigma)$ since $\text{tr}(\Sigma^\dagger)$ or $\text{tr}(\Sigma)$ is not invariant since $\Sigma \rightarrow U_L \Sigma U_R^\dagger$. So the term quadratic in Σ transforms as:

$$Q \equiv \text{tr}(\Sigma^\dagger \Sigma) \rightarrow \text{tr}[(U_L \Sigma U_R^\dagger)^\dagger (U_L \Sigma U_R^\dagger)] \equiv Q' \Rightarrow Q \rightarrow Q'$$

$$\Rightarrow Q' = \text{tr}[(\Sigma U_R^\dagger)^\dagger U_L^\dagger U_L \Sigma U_R^\dagger] \quad \| U_R, U_L \text{ are unitary operators} \Rightarrow U_{L,R} U_{L,R}^\dagger = \mathbb{1}$$

$$Q = \text{tr}[U_R \Sigma^\dagger \Sigma U_R^\dagger]$$

$\|$ Trace of product of matrices is cyclic: $\text{tr}(ABC) = \text{tr}(CAB)$

$$Q' = \text{tr}[U_R^\dagger U_R \Sigma^\dagger \Sigma] = \text{tr}[\Sigma^\dagger \Sigma] = Q$$

$$\Rightarrow Q = Q' \Rightarrow Q \text{ is invariant under symmetry transformation of } SU(2)_L \times SU(2)_R$$

Now let's see if the term cubic in Σ is invariant:

$\Rightarrow C \equiv \text{tr}(\Sigma^\dagger \Sigma \Sigma)$, we can see just from this that this will not be invariant under the transformation since no matter the cubic contribution there will odd number of either $U_{L,R}$ or $U_{L,R}^\dagger$ - which means it's not possible to make all the unitary operators cancel

each other, this means $\text{tr}(\Sigma \Sigma \Sigma)$ or $\text{tr}(\Sigma^\dagger \Sigma \Sigma^\dagger)$ don't work either. Now let's move on to the term quartic in terms of Σ :

$$K \equiv \text{tr}((\Sigma^\dagger \Sigma)^2) \rightarrow K' \equiv \text{tr}[\{(U_L \Sigma U_R^\dagger)^\dagger (U_L \Sigma U_R^\dagger)\}^2]$$

$$\Rightarrow K' = \text{tr}[\{U_R \Sigma^\dagger \Sigma U_R^\dagger\}^2] = \text{tr}[\{U_R \Sigma^\dagger \Sigma \underbrace{U_R^\dagger U_R}_{=1} \Sigma^\dagger \Sigma U_R^\dagger\}]$$

$$= \text{tr}[U_R \Sigma^\dagger \Sigma \Sigma^\dagger \Sigma U_R^\dagger], \quad \parallel \text{Trace is cyclic}$$

$$K' = \text{tr}[U_R^\dagger U_R (\Sigma^\dagger \Sigma)^2] = \text{tr}[(\Sigma^\dagger \Sigma)^2] = K$$

$$\Rightarrow K = K' \Rightarrow K = \text{tr}((\Sigma^\dagger \Sigma)^2) \text{ is invariant under } SU(2)_L \times SU(2)_R$$

Any more powers of $(\Sigma^\dagger \Sigma)$ would have dimensions of over 4 so we cannot have them in our Lagrangian. We can however introduce one more term to Lagrangian and that is

$$\text{tr}(\partial_\mu \Sigma^\dagger \partial^\mu \Sigma) \quad \text{since} \quad [\partial_\mu] = [\partial^\mu] = 1 \quad \parallel \begin{aligned} [dx^\mu] &= -4 \\ \text{in the action } S \\ S &= \int dx^\mu \mathcal{L} \\ \Rightarrow [\partial_\mu] &= [\frac{\partial}{\partial x^\mu}] = 1 \end{aligned}$$

Let's check if this term is invariant under $SU(3)_L \times SU(3)_R$

$$\Rightarrow D \equiv \text{tr}(\partial_\mu \Sigma^\dagger \partial^\mu \Sigma) \rightarrow D' \equiv \text{tr}[\partial_\mu (U_L \Sigma U_R^\dagger)^\dagger \partial^\mu (U_L \Sigma U_R^\dagger)]$$

We are working with global symmetry $SU(2)_L \times SU(2)_R \rightarrow$ The generators $\alpha_{L,R}$ are independent of coordinates.

$$\Rightarrow D' = \text{tr}[U_R (\partial_\mu \Sigma^\dagger) \underbrace{U_L^\dagger U_L}_{=1} (\partial^\mu \Sigma) U_R^\dagger] \quad \parallel \begin{aligned} &\text{Again we use} \\ &\text{the cyclical properties} \\ &\text{of trace to get } U_R^\dagger U_R = \mathbb{1} \end{aligned}$$

$$D' = \text{tr}[(\partial_\mu \Sigma^\dagger)(\partial^\mu \Sigma)] = D$$

$$\Rightarrow D = D' \Rightarrow D = \text{tr}(\partial_\mu \Sigma^\dagger \partial^\mu \Sigma) \text{ is invariant under } SU(2)_L \times SU(2)_R$$

These are the only terms with dimension of 4 or less that we can put in the Lagrangian, since we don't have any interacting fields Σ is coupled to. Now we can construct the Lagrangian

$$\Rightarrow \mathcal{L} = \frac{1}{4} \text{tr}[(\partial_\mu \Sigma^\dagger)(\partial^\mu \Sigma)] - \frac{\mu^2}{4} \text{tr}(\Sigma^\dagger \Sigma) - \frac{\lambda}{16} \text{tr}((\Sigma^\dagger \Sigma)^2),$$

Where $[\mu] = 1$ is dimensionfull coupling constant and $[\lambda] = 0$ is dimensionless coupling constants, the coefficients $\frac{1}{4}$, $\frac{1}{16}$ are chosen to reflect the lecture notes but they are only a matter of conventions.

We can further simplify the $\text{tr}((\Sigma^\dagger \Sigma)^2)$ - term using the following equation for 2×2 - matrix A :

$$\text{tr}(A^2) = [\text{tr}(A)]^2 - 2\det(A)$$

$$\Rightarrow \text{tr}[(\Sigma^\dagger \Sigma)^2] = [\text{tr}(\Sigma^\dagger \Sigma)]^2 - 2\det(\Sigma^\dagger \Sigma)$$

The determinant doesn't contribute to the Lagrangian since it isn't invariant under $SU(2)_L \times SU(2)_R$. Since $\det(\Sigma^\dagger \Sigma) \rightarrow \det(U_L \Sigma^\dagger \Sigma U_R^\dagger)$.

(There may also be another reason for the determinant term contributing and it actually being invariant under the transformation due to some property of determinants that I'm unaware of, it could also be that the dimensionality doesn't meet our condition or that the original $\text{tr}((\Sigma^\dagger \Sigma)^2)$ wasn't actually invariant and the calculation I did was wrong and this determinant is the non-invariant part and $\text{tr}[\Sigma^\dagger \Sigma]$ is the invariant part.)

Anyways the result should just be the following Lagrangian:

$$\mathcal{L} = \frac{1}{4} \text{tr}[\partial_\mu \Sigma^\dagger \partial^\mu \Sigma] - \frac{\mu^2}{4} \text{tr}[\Sigma^\dagger \Sigma] - \frac{\lambda}{16} [\text{tr}(\Sigma^\dagger \Sigma)]^2$$

Additional note: We could've alternatively constructed the Lagrangian with only $[\text{tr}(\Sigma^\dagger \Sigma)]^n$ terms like we do for ϕ^n , e.g. $\phi^4 = (\phi^2)^2$ for complex ϕ and it would consistently and unambiguously lead to the Lagrangian with $[\text{tr}(\Sigma^\dagger \Sigma)]^2$

Exercise 4.7

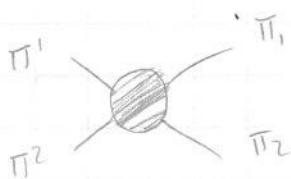
$$\mathcal{L}_1 = \frac{1}{2} (\partial_\mu \sigma \partial^\mu \sigma - 2|\mu|^2 \sigma^2) + \frac{1}{2} \partial_\mu \pi \partial^\mu \pi - \lambda v \sigma (\sigma^2 + \pi \cdot \pi) - \frac{\lambda}{4} (\sigma^2 + \pi \cdot \pi)^2$$

$$\mathcal{L}_2 = \frac{1}{4} (v + \sigma)^2 \text{tr}(\partial_\mu U \partial^\mu U^\dagger)$$

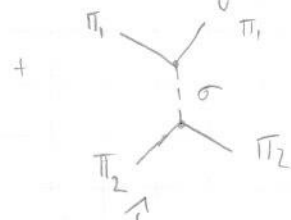
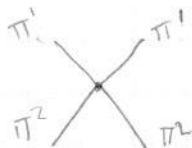
Where $m_\sigma^2 = 2|\mu|^2 = 2\lambda v^2$ and $U = \exp(i\tau \cdot \pi/v)$

We want to calculate

$\pi^1 \pi^2 \rightarrow \pi^1 \pi^2$ at tree level



For $\mathcal{L}_1$ we have 2-diagrams:



This diagram has both s- and t-channel

To calculate the amplitude we need the Feynman rules for the $\pi^1 \pi^2 \pi^1 \pi^2$ - vertex, $\pi^1 \pi^1 \sigma$ - vertex and σ - propagator (Scalar)

to find the Feynman rules. Let's expand some terms of the Lagrangian $\mathcal{L}$.

The $\frac{1}{2} (\partial_\mu \sigma \partial^\mu \sigma - 2\mu^2 \sigma^2)$ -term gives the scalar propagator

$$\frac{i}{p^2 - 2\mu^2}$$

Let's expand $-\lambda v \sigma (\sigma^2 + \pi \cdot \pi)$ to get $\sigma \pi_1 \pi_1$ and $\sigma \pi_2 \pi_2$ vertices:

$$-\lambda v \sigma (\sigma^2 + \pi \cdot \pi) = -\lambda v \sigma^3 - \lambda v \sigma (\pi_1^2 + \pi_2^2 + \pi_3^2)$$

$$= -\lambda v \sigma^3 - \lambda v \sigma \pi_1^2 - \lambda v \sigma \pi_2^2 - \lambda v \sigma \pi_3^2$$

$\Rightarrow$ The Feynman rules for the 3-vertices are:

$$\pi^1 \pi^1 \sigma = -2i\lambda v \quad \text{and} \quad \pi^2 \pi^2 \sigma = -2i\lambda v$$

The coefficient 2 is put here based on the Linear Sigma Feynman rules of Peskin Figure 11.3, this is consistent with our other rules since they have similar form as Peskin does. The coefficient 2 may be explained by the fact that when doing Wick contractions σ can couple to two π^i 's in 2 ways which in our convention would add $\frac{1}{2}$ to the coeff of $-\lambda v \sigma (\pi^i)^2$ which gives $-\frac{2\lambda v}{2} \sigma (\pi^i)^2 = -\frac{1}{2} (2\lambda v) \sigma (\pi^i)^2$
 $\Rightarrow$ rules have $-2i\lambda v$.

Now to get the 4-point vertex we need to expand $-\lambda/4 (\sigma^2 + \pi \cdot \pi)^2$

$$\Rightarrow -\frac{\lambda}{4} (\sigma^2 + \pi \cdot \pi)^2 = -\frac{\lambda}{4} [\sigma^4 + (\pi_1^2 + \pi_2^2 + \pi_3^2)^2 + 2\sigma^2 (\pi_1^2 + \pi_2^2 + \pi_3^2)]$$

$$= -\frac{\lambda}{4} [\sigma^4 + 2\sigma^2 \sum_i (\pi_i)^2 + (\pi_1^4 + \pi_2^4 + \pi_3^4 + 2\pi_1^2 \pi_2^2 + 2\pi_1^2 \pi_3^2 + 2\pi_2^2 \pi_3^2)]$$

$\pi^i \equiv \pi_i$
 and I've been using this quite liberally but index is always lower when power is involved or π^i is in brackets
 $\pi_i^2 = (\pi_i)^2$

We get from this the Feynman rule:

$$\pi_1 \pi_1 \pi_2 \pi_2 = -4i\lambda$$

Now that we have the following Feynman rules:

π_1 π_1
 π_2 π_2

$\pi_1 \quad \pi_1$

$= -2i\lambda v$


 $= -2i\lambda v$

and

$$\frac{\sigma}{p} = \frac{i}{p^2 - 2|\mu^2|}$$

we can finally calculate the amplitude of $\pi, \pi_2 \rightarrow \pi, \pi_2$

S-channel

S-channel

The Feynman rules give us the matrix element:

$$-iM = (-4i\lambda) + (-2i\lambda v) \frac{i}{t - 2|\mu|^2} (-2i\lambda v) + (-2i\lambda v) \frac{i}{s - 2|\mu|^2} (-2i\lambda v)$$

$$\Rightarrow \mu = 4\lambda + \frac{4\lambda^2 v^2}{t - 2|\mu^2|} + \frac{4\lambda^2 v^2}{s - 2|\mu^2|}$$

~~2~~ The amplitude is then

$$|M|^2 = MM^\dagger = \left(4\lambda + \frac{4\lambda^2 v^2}{t - 2|\mu|^2} + \frac{4\lambda^2 v^2}{s - 2|\mu|^2} \right)^2$$

We can expand
this for low momentum
and will do so
later to compare with
result from L_2

→ The amplitude for $\pi_1 \pi_2 \rightarrow \pi_1 \pi_2$ is

$$|M|^2 = \left(4\lambda + \frac{4\pi v^2}{t-2|\mu|^2} + \frac{4\pi^2 v^2}{s-2|\mu|^2} \right)^2$$

Now we calculate

using the second Lagrangian

$$\mathcal{L}_2 = \frac{1}{4} (v + \sigma)^2 \text{tr} (\partial_\mu U \partial^\mu U^\dagger), \quad \text{Let's expand } U \text{ and } U^\dagger$$

$$U = \exp(i \tau \cdot \pi / \nu)$$

$$= \exp(i \frac{\vec{\tau} \cdot \vec{\pi}}{\nu}) = \sum_{k=0}^{\infty} \frac{1}{k!} (i \frac{\vec{\tau} \cdot \vec{\pi}}{\nu})^k$$

$$U = 1 + i \frac{(\vec{r} \cdot \vec{\pi})}{v} - \frac{1}{2} \frac{(\vec{r} \cdot \vec{\pi})^2}{v^2} - \frac{i}{3!} \frac{(\vec{r} \cdot \vec{\pi})^3}{v^3} + \dots$$

|| We need to expand
this up to $k=3$, since
(this will become clear
as we do the calculation)

Let's now calculate $\partial_\mu U$:

$$\partial_\mu U = i \frac{1}{v} \partial_\mu (\vec{\tau} \cdot \vec{\pi}) - \frac{1}{2} \frac{1}{v^2} \partial_\mu [(\vec{\tau} \cdot \vec{\pi})^2] - \frac{i}{3!} \frac{1}{v^3} \partial_\mu [(\vec{\tau} \cdot \vec{\pi})^3]$$

$$\partial_\mu (\vec{\tau} \cdot \vec{\pi}) = \vec{\tau} \cdot \partial_\mu \vec{\pi} \quad \text{Since } \vec{\tau} = (\sigma_1, \sigma_2, \sigma_3) \quad \text{Pauli matrices}$$

$$\Rightarrow \partial_\mu [(\vec{\tau} \cdot \vec{\pi})^2] = \partial_\mu [(\sigma_1 \pi_1 + \sigma_2 \pi_2 + \sigma_3 \pi_3)^2]$$

The Pauli matrices have the following properties:

$$\{\sigma_i, \sigma_j\} = 2 \delta_{ij} \mathbb{1}, \quad \sigma_i^2 = \mathbb{1}, \quad \text{tr}(\sigma_i) = 0$$

$$\begin{aligned} (\sigma_1 \pi_1 + \sigma_2 \pi_2 + \sigma_3 \pi_3)^2 &= \sigma_1^2 \pi_1^2 + \sigma_2^2 \pi_2^2 + \sigma_3^2 \pi_3^2 + \{\sigma_1, \sigma_2\} \pi_1 \pi_2 + \{\sigma_1, \sigma_3\} \pi_1 \pi_3 + \{\sigma_2, \sigma_3\} \pi_2 \pi_3 \\ &= \sum_i \sigma_i^2 \pi_i^2 = \mathbb{1} \sum_i \pi_i^2 = \mathbb{1} \vec{\pi}^2 \end{aligned}$$

$$\Rightarrow \partial_\mu [(\vec{\tau} \cdot \vec{\pi})^2] = \partial_\mu (\mathbb{1} \vec{\pi}^2) = 2 \mathbb{1} \vec{\pi} \cdot \partial_\mu \vec{\pi}$$

The property $\text{tr}(\sigma_i) = 0$ insures that the odd powers of $\vec{\pi}$ or $\partial_\mu \vec{\pi}$ won't contribute to the trace, so we won't get any $\checkmark$ - vertices so we can ignore σ in $\frac{1}{4}(v + \sigma)^2$ for our calculation. Let's continue with calculating the derivative.

$$\begin{aligned} \partial_\mu [(\vec{\tau} \cdot \vec{\pi})^3] &= (\partial_\mu [(\vec{\tau} \cdot \vec{\pi})^2]) (\vec{\tau} \cdot \vec{\pi}) + [\partial_\mu (\vec{\tau} \cdot \vec{\pi})] (\vec{\tau} \cdot \vec{\pi})^2 \\ &= 2 \mathbb{1} (\vec{\pi} \cdot \partial_\mu \vec{\pi}) (\vec{\tau} \cdot \vec{\pi}) + \mathbb{1} (\vec{\tau} \cdot \partial_\mu \vec{\pi}) \vec{\pi}^2 \quad \parallel \mathbb{1} \vec{\pi} = \vec{\pi} \\ &= 2 (\vec{\pi} \cdot \partial_\mu \vec{\pi}) (\vec{\tau} \cdot \vec{\pi}) + (\vec{\tau} \cdot \partial_\mu \vec{\pi}) \vec{\pi}^2 \end{aligned}$$

$$\Rightarrow \partial_\mu U = \frac{i}{v} (\vec{\tau} \cdot \partial_\mu \vec{\pi}) - \frac{1}{2v^2} 2 \mathbb{1} (\vec{\pi} \cdot \partial_\mu \vec{\pi}) - \frac{i}{3!} \frac{1}{v^3} [2 (\vec{\pi} \cdot \partial_\mu \vec{\pi}) (\vec{\tau} \cdot \vec{\pi}) + (\vec{\tau} \cdot \partial_\mu \vec{\pi}) \vec{\pi}^2]$$

$$\partial_\mu U = \frac{i}{v} (\vec{\tau} \cdot \partial_\mu \vec{\pi}) - \frac{1}{v^2} (\vec{\pi} \cdot \partial_\mu \vec{\pi}) - \frac{i}{3!} \frac{1}{v^3} [(\vec{\tau} \cdot \partial_\mu \vec{\pi}) \vec{\pi}^2 + 2 (\vec{\tau} \cdot \vec{\pi}) (\vec{\pi} \cdot \partial_\mu \vec{\pi})]$$

This directly gives us $\partial^\mu U$:

$$\partial^\mu U = -\frac{i}{v} (\vec{\tau} \cdot \partial^\mu \vec{\pi}) - \frac{1}{v^2} (\vec{\pi} \cdot \partial^\mu \vec{\pi}) + \frac{i}{3!} \frac{1}{v^3} [(\vec{\tau} \cdot \partial^\mu \vec{\pi}) \vec{\pi}^2 + 2 (\vec{\tau} \cdot \vec{\pi}) (\vec{\pi} \cdot \partial^\mu \vec{\pi})]$$

We can now calculate $\partial_\mu U \partial^\mu U$ and take the trace.

Since we are only interested in $\pi_1 \pi_2 \rightarrow \pi_1, \pi_2$ and $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$ we can only look at terms that are proportional to $\frac{1}{v^4}$

$$\partial_\mu U \partial^\mu U^\dagger = \left\{ \frac{i}{v} (\vec{\pi} \cdot \partial_\mu \vec{\pi}) - \frac{1}{v^2} (\vec{\pi} \cdot \partial_\mu \vec{\pi}) - \frac{i}{3!} \frac{1}{v^3} [(\vec{\pi} \cdot \partial_\mu \vec{\pi}) \vec{\pi}^2 + 2(\vec{\pi} \cdot \vec{\pi})(\vec{\pi} \cdot \partial_\mu \vec{\pi})] \right\} \\ \cdot \left\{ -\frac{i}{v} (\vec{\pi} \cdot \partial^\mu \vec{\pi}) - \frac{1}{v^2} (\vec{\pi} \cdot \partial^\mu \vec{\pi}) + \frac{i}{3!} \frac{1}{v^3} [(\vec{\pi} \cdot \partial^\mu \vec{\pi}) \vec{\pi}^2 + 2(\vec{\pi} \cdot \vec{\pi})(\vec{\pi} \cdot \partial^\mu \vec{\pi})] \right\}$$

The terms proportional to $\frac{1}{v^4}$ (quartic in the π_i -field) is:

$$i(\vec{\pi} \cdot \partial_\mu \vec{\pi}) \frac{i}{3!} [(\vec{\pi} \cdot \partial^\mu \vec{\pi}) \vec{\pi}^2 + 2(\vec{\pi} \cdot \vec{\pi})(\vec{\pi} \cdot \partial^\mu \vec{\pi})] + \mathbb{1}^2 \underbrace{(\vec{\pi} \cdot \partial_\mu \vec{\pi})(\vec{\pi} \cdot \partial^\mu \vec{\pi})}_{=(\vec{\pi} \cdot \partial_\mu \vec{\pi})^2} \\ - \frac{i}{3!} [(\vec{\pi} \cdot \partial_\mu \vec{\pi}) \vec{\pi}^2 + 2(\vec{\pi} \cdot \vec{\pi})(\vec{\pi} \cdot \partial_\mu \vec{\pi})] (-i)(\vec{\pi} \cdot \partial^\mu \vec{\pi})$$

Now we can again $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{1}$ to write

$$(\vec{\pi} \cdot \vec{A})(\vec{\pi} \cdot \vec{B}) = \vec{A} \cdot \vec{B}$$

Proof: $\vec{\pi} \cdot \vec{A} = (\sigma_1 A_1 + \sigma_2 A_2 + \sigma_3 A_3)$, $\vec{\pi} \cdot \vec{B} = \sigma_1 B_1 + \sigma_2 B_2 + \sigma_3 B_3$

$$\Rightarrow (\vec{\pi} \cdot \vec{A})(\vec{\pi} \cdot \vec{B}) = (\sigma_1 A_1 + \sigma_2 A_2 + \sigma_3 A_3)(\sigma_1 B_1 + \sigma_2 B_2 + \sigma_3 B_3)$$

$$= \sum_i \underbrace{\sigma_i^2}_{=1} A_i B_i + \underbrace{\{\sigma_1, \sigma_2\}}_{=0} A_1 B_2 + \underbrace{\{\sigma_1, \sigma_3\}}_{=0} A_1 B_3 + \underbrace{\{\sigma_2, \sigma_3\}}_{=0} A_2 B_3 \\ = \mathbb{1} \sum_i A_i B_i = \mathbb{1} \vec{A} \cdot \vec{B}$$

Using this our term that is proportional to $\frac{1}{v^4}$ (let's name it ξ) (inside the trace) is

$$\xi = -\frac{\pi}{3!} \left[\underbrace{(\partial_\mu \vec{\pi}) \cdot (\partial^\mu \vec{\pi}) \vec{\pi}^2}_{=(\partial_\mu \vec{\pi})^2} + 2 \underbrace{(\vec{\pi} \cdot \partial_\mu \vec{\pi})(\vec{\pi} \cdot \partial^\mu \vec{\pi})}_{=(\vec{\pi} \cdot \partial_\mu \vec{\pi})^2} \right] + \mathbb{1} (\vec{\pi} \cdot \partial_\mu \vec{\pi})^2$$

$$- \frac{\pi}{3!} \left[\underbrace{(\partial_\mu \vec{\pi}) \cdot (\partial^\mu \vec{\pi}) \vec{\pi}^2}_{=(\partial_\mu \vec{\pi})^2} + 2 \underbrace{(\vec{\pi} \cdot \partial_\mu \vec{\pi})(\vec{\pi} \cdot \partial^\mu \vec{\pi})}_{=(\vec{\pi} \cdot \partial_\mu \vec{\pi})^2} \right]$$

$$\xi = -\frac{2}{3!} \mathbb{1} ((\partial_\mu \vec{\pi})^2 \vec{\pi}^2 + 2(\vec{\pi} \cdot \partial_\mu \vec{\pi})^2) + \mathbb{1} (\vec{\pi} \cdot \partial_\mu \vec{\pi})^2 \quad || \frac{2}{3!} = \frac{1}{3}$$

$$= \mathbb{1} \frac{1}{3} \left(-(\partial_\mu \vec{\pi})^2 \vec{\pi}^2 - (\vec{\pi} \cdot \partial_\mu \vec{\pi})^2 \right) \quad \text{Let's now calculate this}$$

$$(\vec{\pi} \cdot \partial_\mu \vec{\pi})^2 = (\pi_1 \partial_\mu \pi_1 + \pi_2 \partial_\mu \pi_2 + \pi_3 \partial_\mu \pi_3)^2$$

$$= \sum_i \pi_i^2 (\partial_\mu \pi_i)^2 + 2(\pi_1 \partial_\mu \pi_1)(\pi_2 \partial^\mu \pi_2) + 2\pi_1 \pi_3 (\partial_\mu \pi_2)(\partial^\mu \pi_3) + 2\pi_2 \pi_3 \partial_\mu \pi_2 \partial^\mu \pi_3$$

and

$$(\partial_\mu \vec{\pi})^2 \vec{\pi}^2 = (\partial_\mu \pi_1 \partial^\mu \pi_1 + \partial_\mu \pi_2 \partial^\mu \pi_2 + \partial_\mu \pi_3 \partial^\mu \pi_3) (\pi_1^2 + \pi_2^2 + \pi_3^2) \quad \left\| \quad (\partial_\mu \vec{\pi})^2 = (\partial_\mu \pi_i) (\partial^\mu \pi_i) \right.$$

$$= \sum_i \pi_i^2 (\partial_\mu \pi_i)^2 + \pi_1^2 (\partial_\mu \pi_2)^2 + \pi_1^2 (\partial_\mu \pi_3)^2 + \pi_2^2 (\partial_\mu \pi_1)^2 + \pi_2^2 (\partial_\mu \pi_3)^2 + \pi_3^2 (\partial_\mu \pi_1)^2 + \pi_3^2 (\partial_\mu \pi_2)^2$$

We are only interested in terms that are relevant for $\pi_1, \pi_2 \rightarrow \pi_1, \pi_2$.
So we drop terms with π_3 from ξ to get

$$\xi = \frac{1}{3} \mathbb{1} \left[2 \pi_1 \pi_2 (\partial_\mu \pi_1) (\partial^\mu \pi_2) - \pi_1^2 (\partial_\mu \pi_2) (\partial^\mu \pi_2) - \pi_2^2 (\partial_\mu \pi_1) (\partial^\mu \pi_1) \right]$$

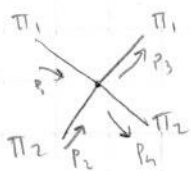
Now we take the trace:

$$\text{tr} \left(\frac{1}{V^4} \xi \right) = \frac{2}{3} \left[2 \pi_1 \pi_2 (\partial_\mu \pi_1) (\partial^\mu \pi_2) - \pi_1^2 \partial_\mu \pi_2 \partial^\mu \pi_2 - \pi_2^2 \partial_\mu \pi_1 \partial^\mu \pi_1 \right] \frac{1}{V^4}$$

The relevant term in Lagrangian is then (we ignore terms with σ)

$$\frac{1}{4} V^2 \text{tr}(\xi) = + \frac{1}{3!} \left[2 \pi_1 \pi_2 (\partial_\mu \pi_1) (\partial^\mu \pi_2) - \pi_1^2 \partial_\mu \pi_2 \partial^\mu \pi_2 - \pi_2^2 \partial_\mu \pi_1 \partial^\mu \pi_1 \right] \frac{1}{V^2}$$

This gives us the following Feynman rules:



$$= i \left(2 p_{1\mu} p_2^\mu - p_{2\mu} p_4^\mu - p_{1\mu} p_3^\mu \right) \frac{1}{V^2}$$

$$= i \left(2 p_1 \cdot p_2 - p_2 \cdot p_4 - p_1 \cdot p_3 \right) \frac{1}{V^2}$$

$$= i \left(s + t_2 + t_1 \right) = i(s+t) \frac{1}{V^2}$$

$$= i(s+t) \frac{1}{V^2} \Rightarrow -i\mathcal{M} = i(s+t) \frac{1}{V^2} \Rightarrow \mathcal{M} = - \frac{s+t}{V^2}$$

Mandelstam variables

for massless π_i 's:

$$\left\| \begin{aligned} -t/2 &= p_1 \cdot p_3 = p_2 \cdot p_4 \\ s/2 &= p_1 \cdot p_2 \end{aligned} \right.$$

$$\Rightarrow \text{The amplitude is the } |\mathcal{M}|_{L_2} = \mathcal{M} \mathcal{M}^\dagger = \underline{\underline{\left[-\frac{s+t}{V^2} \right]^2}}$$

this is the amplitude we get from L_2 , we can now compare this with the amplitude of L_1 :

$$|\mathcal{M}|_{L_1}^2 = \left(4\lambda + \frac{4\lambda^2 v^2}{t-2|\mu^2|} + \frac{4\lambda^2 v^2}{s-2|\mu^2|} \right)^2$$

Let's now expand this for low momentum, where

$$\frac{1}{t-2|\mu^2|} = \frac{-1}{2|\mu^2|} \frac{1}{1-t/2|\mu^2|} = - \frac{1}{2|\mu^2|} \left(1 - \frac{t}{2|\mu^2|} \right)^{-1}$$

$$\left\| \begin{aligned} t &\ll 2|\mu^2| \\ s &\ll 2|\mu^2| \end{aligned} \right.$$

Now we use binomial approximation $(1+x)^{-1} \approx 1-x$, for $x \ll 1$

$$\Rightarrow \frac{1}{t-2|\mu^2|} \approx -\frac{1}{2|\mu^2|} \left(1 + \frac{t}{2|\mu^2|}\right)$$

$$\Rightarrow \frac{1}{s-2|\mu^2|} \approx -\frac{1}{2|\mu^2|} \left(1 + \frac{s}{2|\mu^2|}\right)$$

$$\Rightarrow \frac{4\lambda^2 v^2}{t-2|\mu^2|} \approx -\frac{4\lambda^2 v^2}{2\lambda v^2} \left(1 + \frac{t}{2\lambda v^2}\right) = -2\lambda - \frac{t}{v^2}$$

$$\Rightarrow \frac{4\lambda^2 v^2}{s-2|\mu^2|} \approx -\frac{4\lambda^2 v^2}{2\lambda v^2} \left(1 + \frac{s}{2\lambda v^2}\right) = -2\lambda - \frac{s}{v^2}$$

$$\Rightarrow |M|_{L_1}^2 = (4\lambda - 2\lambda - 2\lambda - \frac{t}{v^2} - \frac{s}{v^2})^2$$

$$|M|_{L_1}^2 = \left(-\frac{t+s}{v^2}\right)^2 = |M|_{L_2}^2$$

So the amplitudes are same for both Lagrangians.

Exercise 4.8

Calculate decay widths of Higgs boson to lepton pair, quark pair, W^+W^- and to ZZ as a function of Higgs mass:

From Peskin 20.6 we get the following Feynman rules:

$$h \text{ --- } \begin{array}{c} \nearrow f \\ \searrow \bar{f} \end{array} = -i \frac{m_f}{v}$$

Higgs $\rightarrow$ fermions

$$\text{here } v = m_h \frac{1}{\sqrt{2}\lambda}$$

$$h \text{ --- } \begin{array}{c} \nu W^+ \\ \mu W^- \end{array} = 2i \frac{m_W^2}{v} g^{\mu\nu}$$

$$h \text{ --- } \begin{array}{c} Z \\ Z \end{array} = 2i \frac{m_Z^2}{v} g^{\mu\nu}$$

The matrix element for $H \rightarrow f\bar{f}$ is

$$\begin{array}{c} p_1 \quad f_s \\ \nearrow \\ \text{---} q \quad \text{---} \\ \searrow \\ p_2 \quad \bar{f}_r \end{array} : -i\mathcal{M} = 1 \cdot \bar{u}_s(p_1) \left(-i \frac{m_f}{v}\right) v_r(p_2)$$

s, r spin indices

For leptons we only have spins, and we sum over them since they only appear in final state

$$|\overline{\mathcal{M}}|^2 = \sum_{s,r} \mathcal{M} \mathcal{M}^\dagger = \sum_{s,r} \bar{u}_s(p_1) \frac{m_f}{v} v_r(p_2) \bar{v}_r(p_2) \frac{m_f}{v} u_s(p_1)$$

Let's write the spinors in component form with sum over repeated indices:

$$|\bar{\mathcal{M}}|^2 = \sum_{s,r} \frac{m_f^2}{v^2} \bar{u}_a^s(p_1) v_a^r(p_2) \bar{v}_b^s(p_2) u_b^r(p_1)$$

$$= \left(\frac{m_f}{v}\right)^2 \sum_{s,r} u_b^s(p_1) \bar{u}_a^s(p_1) v_a^r(p_2) \bar{v}_b^r(p_2)$$

Now we use the famous spin sum formula

$$\sum_s u^s(p_1) \bar{u}^s(p_1) = \not{p}_1 + m_f, \quad \sum_r v^r(p_2) \bar{v}^r(p_2) = \not{p}_2 - m_f$$

$$\Rightarrow |\bar{\mathcal{M}}|^2 = \left(\frac{m_f}{v}\right)^2 \left(\sum_s u^s(p_1) \bar{u}^s(p_1)\right)_{ba} \left(\sum_r v^r(p_2) \bar{v}^r(p_2)\right)_{ab} \quad \left\| \begin{array}{l} A_{ab} B_{ba} = \text{tr}(AB) \end{array} \right.$$

$$|\bar{\mathcal{M}}|^2 = \left(\frac{m_f}{v}\right)^2 \text{tr}((\not{p}_1 + m_f)(\not{p}_2 - m_f))$$

$$= \left(\frac{m_f}{v}\right)^2 \text{tr}(\not{p}_1 \not{p}_2 - \not{p}_1 m_f + \not{p}_2 m_f - \mathbb{1} m_f^2)$$

$$= \left(\frac{m_f}{v}\right)^2 [\text{tr}(\not{p}_1 \not{p}_2) - 4m_f^2]$$

$$= \left(\frac{m_f}{v}\right)^2 4(p_1 \cdot p_2 - m_f^2) = 4\left(\frac{m_f}{v}\right)^2 \left(\frac{s}{2} - 2m_f^2\right)$$

$$|\bar{\mathcal{M}}|_e^2 = 2\left(\frac{m_f}{v}\right)^2 (s - 4m_f^2) \quad \text{for leptons } f, \bar{f}$$

$$\mathbb{1} = \mathbb{1}_{4 \times 4} = \mathbb{1}_4$$

$$\Rightarrow \text{tr}(\mathbb{1}) = 4$$

$$\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$$

$$\text{tr}(\text{odd \# of } \gamma\text{'s}) = 0$$

$$\Rightarrow \text{tr}(\not{0}) = 0$$

$$\text{tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu}$$

$$p_1 \cdot p_2 = \frac{s}{2} - m_f^2$$

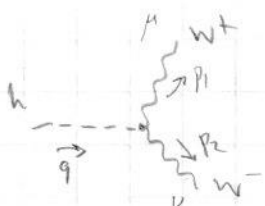
$\uparrow$ Mandelstam variable s

For quarks we also have 3 colours (anti quark will have anti-colour so that quark + antiquark is colour-neutral). When we sum over the three colours we just get factor of 3 since there are three colours.

So we just get

$$|\bar{\mathcal{M}}|_q^2 = 3 |\bar{\mathcal{M}}|_e^2 = 6 \left(\frac{m_f}{v}\right)^2 (s - 4m_f^2) \quad \text{for quarks.}$$

Now let's calculate the amplitude for $W^+ W^-$ decay product



$$-i\mathcal{M} = \epsilon_\mu^*(p_1) \epsilon_\nu^*(p_2) (2i \frac{m_W^2}{v} g^{\mu\nu})$$

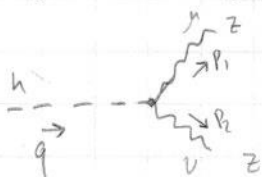
$$\Rightarrow \mathcal{M}\mathcal{M}^\dagger = \epsilon_\mu^*(p_1) \epsilon_\mu(p_1) \epsilon_\nu^*(p_2) \epsilon_\nu(p_2) \frac{4m_W^4}{v^2} g^{\mu\nu} g^{\sigma\rho}$$

We sum over polarizations using $\sum_{\mu=1}^3 \epsilon_\mu(p_1) \epsilon_\mu^*(p_1) = -\left(g_{\mu\mu} - \frac{p_{1\mu} p_{1\mu}}{m_W^2}\right)$ from Peskin eq. (21.26). This is the polarization sum for massive gauge boson with mass m_W and momentum p_1 .

$$\begin{aligned}
 \Rightarrow |\bar{\mathcal{M}}|_{W^+W^-} &= \left(\sum_{\mu=1}^3 \epsilon_\mu^*(p_1) \epsilon_\mu(p_1)\right) \left(\sum_{\nu=1}^3 \epsilon_\nu^*(p_2) \epsilon_\nu(p_2)\right) \frac{4m_W^4}{v^2} g^{\mu\nu} g^{\sigma\rho} \\
 &= \left[g_{\mu\rho} - \frac{p_{1\mu} p_{1\rho}}{m_W^2}\right] \left[g_{\nu\sigma} - \frac{p_{2\nu} p_{2\sigma}}{m_W^2}\right] \frac{4m_W^4}{v^2} g^{\mu\nu} g^{\sigma\rho} \\
 &= \left[\underbrace{g_{\mu\rho} g^{\mu\nu}}_{\delta_\rho^\nu} - \frac{p_1^\nu p_{1\rho}}{m_W^2}\right] \left[\underbrace{g_{\nu\sigma} g^{\sigma\rho}}_{\delta_\nu^\rho} - \frac{p_{2\nu} p_2^\rho}{m_W^2}\right] \frac{4m_W^4}{v^2} \\
 &= \frac{4m_W^4}{v^2} \left[\underbrace{\delta_\rho^\nu \delta_\nu^\rho}_{\delta_\rho^\rho} - \frac{p_2^2}{m_W^2} - \frac{p_1^2}{m_W^2} + \frac{(p_1 \cdot p_2)(p_1 \cdot p_2)}{m_W^4} \right] \quad \left\| \begin{array}{l} \delta_\rho^\rho = 4 \\ p_1^2 = m_W^2 \\ p_2^2 = m_W^2 \end{array} \right. \\
 &= \frac{4m_W^4}{v^2} \left[4m_W^2 - 2m_W^2 + \frac{(p_1 \cdot p_2)^2}{m_W^2} \right] \quad \left\| (p_1 \cdot p_2) = s/2 - m_W^2 \right. \\
 &= \frac{4}{v^2} \left[2m_W^4 + \left[\frac{s}{2} - m_W^2\right]^2 \right]
 \end{aligned}$$

$$|\bar{\mathcal{M}}|_{W^+W^-} = \frac{1}{v^2} \left[8m_W^4 + (s - 2m_W^2)^2 \right]$$

Let's now calculate the amplitude for $h \rightarrow Z Z$



$$: -i\mathcal{M} \epsilon_\mu^*(p_1) \epsilon_\nu^*(p_2) \left(2i \frac{m_Z^2}{v}\right) g^{\mu\nu}$$

$\Rightarrow \mathcal{M}$ of $h \rightarrow ZZ$ is identical to $\mathcal{M}$ of $h \rightarrow W^+W^-$ with m_W replaced by m_Z . This means that the amplitude will also be identical with m_W replaced by m_Z , since both are massive gauge bosons and as such polarization sum is identical (with $m_W \rightarrow m_Z$) so we can use result from $h \rightarrow W^+W^-$

$$\Rightarrow |\bar{\mathcal{M}}|_{ZZ} = \frac{1}{v^2} \left[8m_Z^4 + (s - 2m_Z^2)^2 \right]$$

Now that we have all the amplitudes we can calculate the decay widths using the following formula:

$$\Gamma(x \rightarrow ab) = \frac{1}{2\sqrt{s}} \int d\pi_a d\pi_b (2\pi)^4 \delta^{(4)}(\sqrt{s} - p_a - p_b) |\bar{\mathcal{M}}_{x \rightarrow ab}|^2$$

, where $d\Pi_i = \frac{d^3 p_i}{2E_i (2\pi)^3}$

Since all of our amplitudes are independent of momentum p_i (only depend on s) we can take the amplitude out of the phase-space integral.

$$\Rightarrow \Gamma(x \rightarrow ab) = \frac{|\bar{M}_{x \rightarrow ab}|^2}{2\sqrt{s}} \int \frac{d^3 p_a}{2E_a (2\pi)^3} \frac{d^3 p_b}{2E_b (2\pi)^3} (2\pi)^4 \delta^{(4)}(\sqrt{s} - p_a - p_b)$$

Let's start with writing

$$\delta^{(4)}(\sqrt{s} - p_a - p_b) = \delta(\sqrt{s} - E_a - E_b) \delta^{(3)}(-\vec{p}_a - \vec{p}_b)$$

$$\Rightarrow \Gamma(x \rightarrow ab) = \frac{|\bar{M}_{x \rightarrow ab}|^2}{2\sqrt{s}} \int \frac{d^3 p_a}{2E_a (2\pi)^3} \frac{d^3 p_b}{2E_b (2\pi)^3} (2\pi)^4 \delta(\sqrt{s} - E_a - E_b) \delta^{(3)}(-\vec{p}_a - \vec{p}_b)$$

Let's now integrate over $d^3 p_b$, which sets $\vec{p}_b = -\vec{p}_a$

$$\Gamma(x \rightarrow ab) = \frac{|\bar{M}_{x \rightarrow ab}|^2}{2\sqrt{s}} \int \frac{d^3 p_a}{2E_a (2\pi)^3} \frac{(2\pi)^4}{2E_b (2\pi)^3} \delta(\sqrt{s} - E_a - E_b) \quad \parallel E_a^2 = \vec{p}_a^2 + m_a^2$$

We can write this as a spherical integral since E_a depends on $|\vec{p}|$

$$\Rightarrow \Gamma(x \rightarrow ab) = \frac{|\bar{M}_{x \rightarrow ab}|^2}{2\sqrt{s}} \underbrace{4\pi}_{=\int d\Omega} \int \frac{d|\vec{p}_a|}{2E_a (2\pi)^3} \frac{(2\pi)^4 |\vec{p}_a|^2}{2E_b (2\pi)^3} \delta(\sqrt{s} - E_a - E_b)$$

$$\text{We can now write } \delta(\sqrt{s} - E_a - E_b) = \delta(\sqrt{s} - \sqrt{|\vec{p}_a|^2 + m_a^2} - \sqrt{|\vec{p}_a|^2 + m_b^2})$$

Since the processes we are looking for have $m_a = m_b$, we can simplify before proceeding further: (We have $E_a = E_b$ since mass identical)

$$\delta(\sqrt{s} - E_a - E_b) = \delta(\sqrt{s} - 2\sqrt{|\vec{p}_a|^2 + m_a^2})$$

, now let's use the following property of delta function:

$$\delta(g(x)) = \frac{\delta(x - x_0)}{|g'(x_0)|} \quad \text{where } x_0 \text{ is root of } g(x)$$

we have:

$$g(x) = g(|\vec{p}_a|) = \sqrt{s} - 2\sqrt{|\vec{p}_a|^2 + m_a^2}, \text{ which has root at}$$

$$|\vec{p}_a|^2 = \frac{s}{4} - m_a^2 \quad \Rightarrow \quad |\vec{p}_a| = \sqrt{\frac{s}{4} - m_a^2}$$

$$\Rightarrow \delta(x - x_0) = \delta(|\vec{p}_a| - \sqrt{\frac{s}{4} - m_a^2}) \quad \text{and} \quad g' = -\frac{2|\vec{p}_a|}{\sqrt{|\vec{p}_a|^2 + m_a^2}}$$

$$\Rightarrow g'(x_0) = -\frac{2(\sqrt{\frac{s}{4} - m_a^2})}{\sqrt{(\frac{s}{4} - m_a^2) + m_a^2}} = -\frac{2x_0}{\sqrt{s/4}} = -\frac{4x_0}{\sqrt{s}}$$

$$\Rightarrow \delta(g(x)) = \delta(|p_a| - \sqrt{\frac{s}{4} - m_a^2}) \cdot \frac{\sqrt{s}}{4} \cdot \frac{1}{x_0} \quad || x_0 = \sqrt{\frac{s}{4} - m_a^2}$$

$$\Rightarrow \Gamma(X \rightarrow ab) = \frac{|\overline{\mathcal{M}}_{X \rightarrow ab}|^2}{2\sqrt{s}} 4\pi \int \frac{d|p_a|}{4E_a^2 (2\pi)^2} \frac{\sqrt{s}}{4} \frac{|p_a|^2}{x_0} \delta(|p_a| - \sqrt{\frac{s}{4} - m_a^2})$$

Let's now integrate

$$\begin{aligned} \Gamma(X \rightarrow ab) &= \frac{|\overline{\mathcal{M}}_{X \rightarrow ab}|^2 \pi}{8 (2\pi)^2 E_a^2} \sqrt{\frac{s}{4} - m_a^2} \\ &= \frac{|\overline{\mathcal{M}}_{X \rightarrow ab}|^2}{16\pi (s/4)} \sqrt{\frac{s}{4} - m_a^2} = \frac{|\overline{\mathcal{M}}_{X \rightarrow ab}|^2}{4\pi s} \sqrt{\frac{s}{4} - m_a^2} \end{aligned}$$

$$\begin{aligned} S &= (E_a + E_b)^2 = (2E_a)^2 \\ &= 4E_a^2 \Rightarrow E_a^2 = s/4 \\ E_a^2 - m_a^2 &= |p_a|^2 \end{aligned}$$

Since we have higgs decay $S = m_h^2$

$$\Rightarrow \Gamma(h \rightarrow ab) = \frac{\sqrt{\frac{m_h^2}{4} - m_a^2}}{4\pi m_h^2} |\overline{\mathcal{M}}_{h \rightarrow ab}|^2 = \frac{\sqrt{m_h^2 - 4m_a^2}}{8\pi m_h^2} |\overline{\mathcal{M}}_{h \rightarrow ab}|^2$$

Now let's calculate $h \rightarrow \ell \bar{\ell}$

$$\begin{aligned} \Gamma(h \rightarrow \ell \bar{\ell}) &= \frac{\sqrt{m_h^2 - 4m_\ell^2}}{8\pi m_h^2} |\overline{\mathcal{M}}_\ell|^2 \quad || \begin{aligned} S &= m_h^2 \\ |\overline{\mathcal{M}}_\ell|^2 &= 2\left(\frac{m_\ell}{v}\right)^2 (s - 4m_\ell^2) \end{aligned} \\ &= \frac{\sqrt{m_h^2 - 4m_\ell^2}}{8\pi m_h^2} \left(2 \frac{m_\ell^2}{v^2} (m_h^2 - 4m_\ell^2)\right) \\ &= \frac{(m_h^2 - 4m_\ell^2)^{3/2}}{4\pi m_h^2} \frac{m_\ell^2}{m_h^2} 2\lambda \quad || v = m_h \frac{1}{\sqrt{2}\lambda}, s = m_h^2 \\ \Gamma(h \rightarrow \ell \bar{\ell}) &= \frac{(m_h^2 - 4m_\ell^2)^{3/2}}{2\pi} \frac{\lambda m_\ell^2}{m_h^4} \end{aligned}$$

Since for quark we had $|\overline{\mathcal{M}}|_q^2 = 3|\overline{\mathcal{M}}|_\ell^2$ we get

$$\Gamma(h \rightarrow q \bar{q}) = 3 \frac{(m_h^2 - 4m_q^2)^{3/2}}{2\pi} \frac{\lambda m_q^2}{m_h^4}$$

Note that these two decay widths are for a single flavour with mass m_ℓ or m_q to get the width for any lepton we would have to add decay to individual leptons and same for quarks:

$$\Gamma(h \rightarrow \text{any lepton pair}) = \sum_i \Gamma(h \rightarrow \ell_i \bar{\ell}_i) \quad \text{where } i \text{ is flavor, e.g. } \ell_1 = e = \text{electron}$$

similarly:

$$\Gamma(h \rightarrow \text{any quark pair}) = \sum_i \Gamma(h \rightarrow q_i \bar{q}_i)$$

Now let's calculate the decay width for $h \rightarrow W^+ W^-$, $h \rightarrow Z Z$

$$\begin{aligned}\Gamma(h \rightarrow W^+ W^-) &= \frac{\sqrt{m_h^2 - 4m_W^2}}{8\pi m_h^2} |\bar{M}|_{W^+ W^-}^2 \\ &= \frac{(m_h^2 - 4m_W^2)^{1/2}}{8\pi m_h^2} \frac{1}{v^2} [8m_W^4 + (s - 2m_W^2)^2] \bigg|_{v = \frac{m_h}{\sqrt{2}\lambda}} \\ &= \frac{(m_h^2 - 4m_W^2)^{1/2}}{8\pi m_h^2} \frac{2\lambda}{m_h^2} (8m_W^4 + (m_h^2 - 2m_W^2)^2)\end{aligned}$$

$$\underline{\underline{\Gamma(h \rightarrow W^+ W^-)}} = \frac{(m_h^2 - 4m_W^2)^{1/2}}{4\pi} \frac{\lambda}{m_h^4} (8m_W^4 + (m_h^2 - 2m_W^2)^2) \quad \square$$

The amplitude for $h \rightarrow Z Z$ is same as for $h \rightarrow W^+ W^-$ with m_W replaced by m_Z . So we get decay width directly from $\Gamma(h \rightarrow W^+ W^-)$

$$\Rightarrow \underline{\underline{\Gamma(h \rightarrow Z Z)}} = \frac{(m_h^2 - 4m_Z^2)^{1/2}}{4\pi} \frac{\lambda}{m_h^4} (8m_Z^4 + (m_h^2 - 2m_Z^2)^2)$$

We have computed all of the decay width.